AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2016

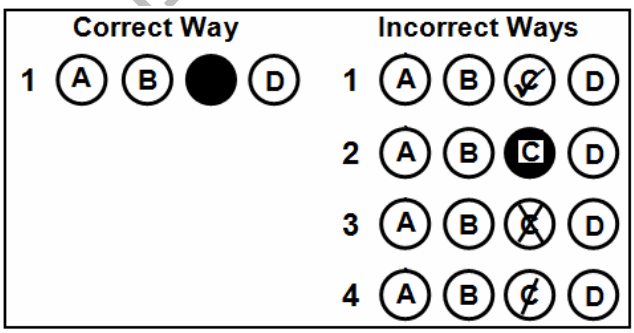
General Mathematics Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. Which of the following algebraic expressions is a polynomial?

   A. $4x^{-2} + 4x + 4$
   B. $2\sqrt{x} + x + 1$
   C. $\dfrac{2x^3 + x^2 + 1}{x}$
   D. $\dfrac{4x^4 - x^3 - 7x}{2x}$

2. The simplest form of $\dfrac{x-y}{x+y} \div \dfrac{x-y}{x+y}$ is

   A. 0
   B. 1
   C. $\dfrac{(x-y)^2}{(x+y)^2}$
   D. $\dfrac{x^2 - y^2}{x^2 + y^2}$

3. If $P(x) = 4x^2 - 2x - 1$, then $P(-2)$ is equal to

   A. 19
   B. 11
   C. -13
   D. -21

4. The factors of $(a+b)^2 - 4$ are

   A. $(a+b+2)(a+b-2)$
   B. $(a+b+4)(a+b-4)$
   C. $(a-b+4)(a+b-4)$
   D. $(a-b+2)(a+b-2)$

5. On complete factorisation of $2x + 3y + 3x + 2y$, we get

   A. $(x+y)\times 3$
   B. $(x+y)\times 2$
   C. $(x+y)\times 5$
   D. $(x+y)\times 6$

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6. On factorisation of $x^2 + 4x + 4$, we get

A. $(x+1)(x+4)$
B. $(x+2)(x+2)$
C. $(x+4)(x+4)$
D. $4(x+1)(x+1)$

7. The least common multiple of (L.C.M.) of $2ab$ and $4b$ is

A. $2b$
B. $4b$
C. $2ab$
D. $4ab$

8. The highest common factor (H.C.F.) of $a + b$ and $a^2 + 2ab + b^2$ is

A. 1
B. $a+b$
C. $(a+b)^2$
D. $(a+b)^3$

9. $\sqrt{\frac{8(a+b)^3}{2(a+b)}}$ is equal to

A. $2(a+b)^2$
B. $4(a+b)^2$
C. $2(a+b)$
D. $4(a+b)$

10. For $\frac{a}{b}x = a$, the value of x is equal to

A. b
B. $-b$
C. $-\frac{a^2}{b}$
D. $\frac{a^2}{b}$

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11. The solution set of $x-\sqrt{a}=\sqrt{a}$ is

A. $\{0\}$
B. $\{a\}$
C. $\{\sqrt{2a}\}$
D. $\{2\sqrt{a}\}$

12. If $x \in Z$, the solution set of $2x<-8$ is

A. $\{-4\}$
B. $\{4,5,6,...\}$
C. $\{...,-7,-6,-5,-4\}$
D. $\{...,-7,-6,-5\}$

13. The solution set of $x^2+6x+9=0$ is

A. $\{3\}$
B. $\{-3\}$
C. $\{-3,3\}$
D. $\{0,-3\}$

14. For the equation $ax^2+bx+c=0$, the value of x will be

A. $x=\dfrac{-b\pm\sqrt{4ac-b^2}}{2a}$
B. $x=\dfrac{b\pm\sqrt{4ac-b^2}}{2b}$
C. $x=\dfrac{b\pm\sqrt{b^2-4ac}}{2b}$
D. $x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$

15. Which of the following matrices is a square matrix?

A. $\begin{bmatrix}1\\1\end{bmatrix}$
B. $\begin{bmatrix}2 & 2\end{bmatrix}$
C. $\begin{bmatrix}1 & 3\\2 & 4\end{bmatrix}$
D. $\begin{bmatrix}1 & 2 & 3\\1 & 2 & 3\end{bmatrix}$

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16. The transpose of the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

A. $\begin{bmatrix} a & d \\ b & c \end{bmatrix}$

B. $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$

C. $\begin{bmatrix} a & c \\ d & b \end{bmatrix}$

D. $\begin{bmatrix} b & c \\ d & a \end{bmatrix}$

17. The determinant of the matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ is

A. 0
B. $2a-2b$
C. a^2-b^2
D. a^2+b^2

18. $\begin{matrix} 2x-3y=-1 \\ 3x+3y=5 \end{matrix}$ can be written in the matrix form as

A. $\begin{bmatrix} 2 & -3 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} -1 \\ 5 \end{bmatrix}$

B. $\begin{bmatrix} 2 & -2 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 1 \\ -5 \end{bmatrix}$

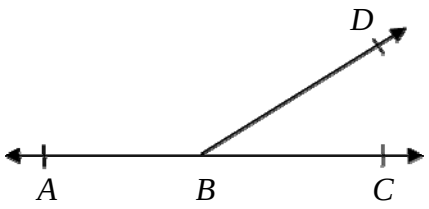
C. $\begin{bmatrix} 2 & 3 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} -1 \\ 5 \end{bmatrix}$

D. $\begin{bmatrix} 2 & 3 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 1 \\ -5 \end{bmatrix}$

19. In the given diagram, $\angle ABD$ and $\angle CBD$ are

I. adjacent angles.
II. supplementary angles.
III. complementary angles.

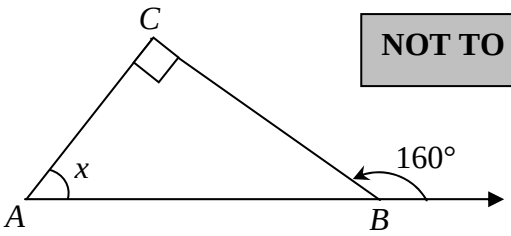
A. I only
B. II only
C. I and II
D. I and III



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20. In the given diagram, the value of x is

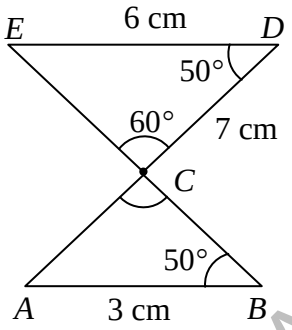
A. 20°
B. 45°
C. 50°
D. 70°



NOT TO SCALE

21. In the given figure, $m\overline{BC}$ is equal to

A. 3 cm
B. 3.5 cm
C. 4 cm
D. 6 cm

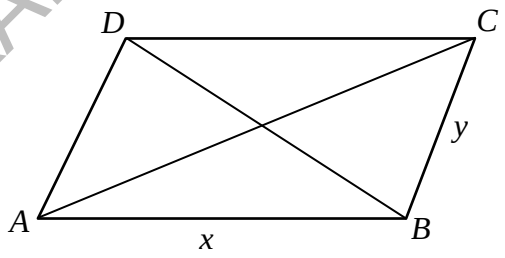


NOT TO SCALE

22. For the given parallelogrm $ABCD$, which of the following options is TRUE?

I. $m\overline{AC} > x$
II. $m\overline{AC} > y$
III. $m\overline{AC} > x + y$

A. I only
B. II only
C. I and II
D. II and III



23. In a triangle, the line segment joining the midpoint of a side to the opposite vertex is called

A. altitude.
B. median.
C. right bisector.
D. angle bisector.

24. If two sides of the right angled triangle are 12 cm and 15 cm, then the third side will be

A. 7 cm
B. 8 cm
C. 9 cm
D. 10 cm

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25. The area of the semi-circle with diameter 12 cm is

A. 18π cm^2
B. $18\pi^2$ cm^2
C. 36π cm^2
D. $36\pi^2$ cm^2

26. The area of an equilateral triangle having each side of measure 2 m is

A. 4 m^2
B. $2\sqrt{3}$ m^2
C. $\sqrt{6}$ m^2
D. $\sqrt{3}$ m^2

27. The radius of a ball is 3 cm, the volume of the ball is

A. 36π cm
B. 36π cm^3
C. 12π cm
D. 12π cm^3

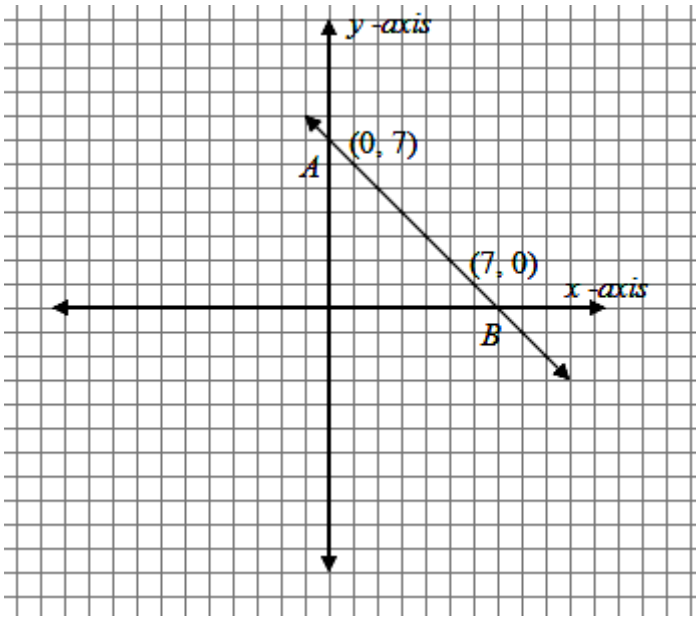
28. The length of the given line segment AB is



A. 0
B. 2
C. 4
D. $2\sqrt{2}$

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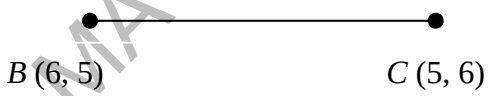
29.



In the above graph, the distance between the points *A* and *B* is

A. 7
B. 14
C. $\sqrt{49}$
D. $\sqrt{98}$

30. The midpoint of the given line segment *BC* is



A. (0.5, 0.5)
B. (5.5, 5.5)
C. (6.5, 6.5)
D. (11, 11)

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